F256 Axial Compensated Loadcell

Geometry: Beam and diaphragm combination. Tension, compression and bi-directional options are available. All standard bi-directional loadcells are calibrated in both modes.

The loadcell's unique strain system compensates for typical force misalignment in force measurement rigs and industrial weighing systems. Maximum error in axial force component measurement is limited to 0.25% within a 3° angle swept through 360° around the loadcell axis. Its various end fixing options are all inert and easily modified for direct inclusion in mechanical assemblies.

The basic versions are all sealed to IP65 with IP67 available as an option. See Engineering Application Sheet E032 and the ICA6H data-sheet for more information.

We are happy to design variants of this loadcell to meet your specific requirements. Versions can be manufactured for fully compensated operation up to +250°C. Please consult our engineering department.

A special version of the F256 loadcell widely used in Formula 1 racing is the Set-up wheel loadcell.

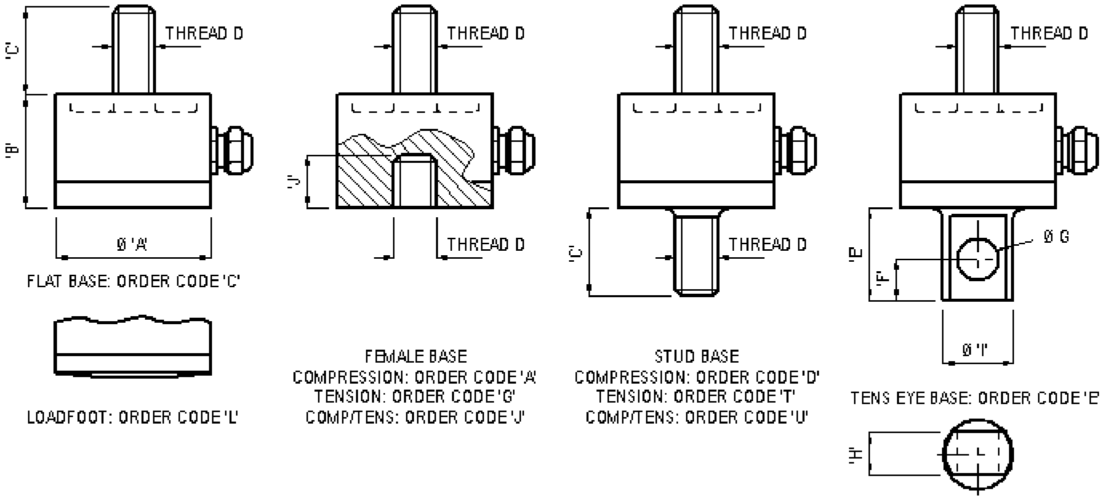
Features

- High accuracy
- Misalignment error compensation
- Highly adaptable inert end fixings
- Standard 1 year warranty
- Output rationalised to 2mV/V
- Traceable calibration with certificate included in the standard price

Tel: 01424852744
Email: info@novatechloadcells.co.uk
Address: 83 Castleham Road, St Leonards on Sea, East Sussex, TN38 9NT, England.

Outline

LOADCELL SIZE	Ø 'A'	'B'	'C'	THREAD 'D'	'E'	'F'	Ø 'G'	'H'	'I'	'J'
SIZE 1 200N - 5kN	44	32	25	M12 x 1.75	26.5	12	12	12	20	15
SIZE 2 10kN - 60kN	66	45	35	M24 x 2	51.5	24	24	24	40	25



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Specifications

Parameter	Value	Unit
Non-linearity - Terminal	±0.05	% RL
Hysteresis	±0.05	% RL
Creep - 20 minutes	±0.05	% AL
Repeatability	±0.02	% RL
Rated output - Rationalised	2.0	mV/V
Rationalisation tolerance (applies to single direction calibrations)	±0.1	% RL
Output symmetry	±0.3	%AO
Zero load output	±4	% RL
Temperature effect on rated output per °C	±0.005	% AL
Temperature effect on zero load output per °C	±0.005	% RL
Temperature range - Compensated	-10 to +50	°C
Temperature range - Safe	-10 to +80	°C
Excitation voltage - Recommended	10	V
Excitation voltage - Maximum	20	V
Bridge resistance	700	Ω
Insulation resistance - Minimum at 50Vdc	500	MΩ
Inclined load error - concentric at 3°	±0.25	% RL
Overload - Safe	50	% RL
Overload - Ultimate	100	% RL
Sealing - R option	IP65	
Sealing - S option	IP67	
Weight - Nominal (T version excluding cable)	200-800N 0.1	kg
	1.25-5kN 0.3	kg
	10-60kN 1.0	kg

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Order Codes

Code	Description
F256CFR0KN	Compression, flat base, IP65
F256LFR0KN	Compression, convex base, IP65
F256TFR0KN	Tension, stud base, IP65
F256DFR0KN	Compression, stud base, IP65
F256UFR0KN	Bi-directional, stud base, IP65
F256EFR0KN	Tension, eye base, IP65
F256AFR0KN	Compression, female base, IP65
F256GFR0KN	Tension, female base, IP65
F256JFR0KN	Bi-directional, female base, IP65
All F256s are rationalised as standard. Change R to an S for IP67. Integral amplifiers are not available with A, G or J options in the standard body height. We can manufacture specials with increased height if integral amplifiers are required.	

Structural Stiffness

Range (kN)	Stiffness (N/m)
200 (N)	7.8×10^6
400 (N)	2.3×10^7
800 (N)	1.2×10^7
1.25	1.9×10^7
2.5	3.9×10^7
5	7.8×10^7
10	1.0×10^8
20	2.0×10^8
40	4.0×10^8
60	6.0×10^8

Notes

- AL = Applied load.
- RL = Rated load.
- Temperature coefficients apply over the compensated range.
- AO=Average of tension and compression outputs for full load.

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Connections

For ranges up to 5kN the loadcell is fitted with 2 metres of PVC insulated 4 core screened cable type 7-2-4C. Ranges above 5kN are fitted with 16-2-4C cable.

Excitation + = Red, Excitation – = Blue, Signal + = Yellow, Signal – = Green, Screen = Orange.

Reverse the signal connections to obtain a positive signal in tension mode. The screen is not connected to the loadcell body.

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